

C-1/W-1(Q-1)

moxug 7my

22/06/26

"Books" $\sim$ sim

"Very Easy" $\sim$ sim

Chap - 1 (Quiz - 1)

Data Mining: Extracting important (and interesting) ^{pattern on} information from any data set

Describe the essential steps of KDD:

* Knowledge Discovery from Data (KDD):

Extracting knowledge from data (synonym of Data Mining)

Steps (sequential):

(i) $\rightarrow$ Data Cleaning: to remove noise and inconsistent data

(ii) $\rightarrow$ Data Integration: ^{where} multiple data sources may be combined
 $\rightarrow$ fixing format may be needed

(iii) $\rightarrow$ Data Selection: data relevant to the analysis are retrieved
(and discard what may not be needed)

(iv) $\rightarrow$ Data Transformation: data are transformed and consolidated into specific forms using (summary and aggregation) $\rightarrow$ format

(v) $\rightarrow$ Data Mining: intelligent methods are applied to extract data patterns

(vi) Pattern Evaluation: identify which patterns are more interesting

(vii) Knowledge Presentation: visualization and representation of findings

(ch-5)

[] Data Warehouse: NOT Database

~~"A"~~ ~~sto~~ A huge ~~storage~~ repository of data
of a large timeline (DB is short time) ↳ say server

C-2, W-1 (Q-1)

24/06/26

What kinds of patterns can be mined

*Characterization and discrimination:

Characterization → similar characteristics between data
→ identifying common characteristics between groups

Discrimination → comparison between two different groups

Mining frequent patterns, associations and correlations

repetitive pattern in a relatively time/instances

foreshadow

connection between items



quantitative strength of association (-1 to 1) *en close enough, welcome back statistics*

support → % of ~~the~~ analysis among all that supports this case
confidence → chance of association (consider left)

✓ *frequent itemset
↳ set of frequent items
↳ same transaction purchase

*frequent subsequence
↳ set of frequent items in different transaction

a frequent substructure
↳ structure of frequent items

support will always be low since there will be a lot of transactions. Meanwhile, confidences are basically probability

support means how many times an association comes up
confidence is how many times among those associations this occurs.

C-3, W-1 (Q-1)

25/06/26

* Class Attribute $\rightarrow$ Feature/Attr that characterizes

* Training $\rightarrow$ has class attr. value

Testing $\rightarrow$ does not have class attr. value

* Outlier $\rightarrow$ Extreme values $\rightarrow$ foreshadow

Chap - 2

Getting to Know Your Data

Data Object and Attribute Types:

* Data Object: a data entity $\rightarrow$ instances

* Attribute: characteristics of a data object (column/feature)
 $\rightarrow$ called dimension in ~~DATA~~ DM

* Set of ^{all} ~~Attributes~~ ^{name} ~~Attributes~~ ^{are} called Attribute Vectors

$\rightarrow$ Nominal Attribute: some sort of name or category (but not ordered)

$\rightarrow$ Binary Attribute: any attribute with two values

$\rightarrow$ Ordinal Attribute: can be ranked $\rightarrow$ (high, medium, low) etc
 $\rightarrow$ required values have high magnitude

$\rightarrow$ Numeric

→ Numeric Attributes : Quantitative attributes (numbers)

→ Interval-Scaled; i) Does not have any true 0 point → $0 \neq \text{empty}$

→ Ratio-Scaled; i) True 0 point

02 C-4, W-2 (Q-2)

02/07/26

49 Central Tendency: which values are more "recurring"

→ Mean, Median, Mode

* Mean: Basic mean / Arithmetic mean =

$$\text{Weighted Mean, } \bar{x} = \frac{\sum w_i x_i}{\sum w_i}$$

* Normal Median

* Median: i) Sort (low to high)

ii) Count if (odd or even) → for even:
if N is numbers of data,
then take,

$$\frac{N-2}{2} \text{ and } \left(\frac{N-2}{2} + 1\right)$$

and take their average

* Interval Median: for binned data (

$$\text{median} = L_i + \left(\frac{\frac{N}{2} - (\sum \text{freq})_L}{\text{freq}_{\text{median}}} \right) \times \text{width} ; L_i = \text{Lower bound of median interval}$$

↳ ascending sorted

median interval: class that contains the $\frac{N}{2}$ th value

N = numbers of values of entire data set

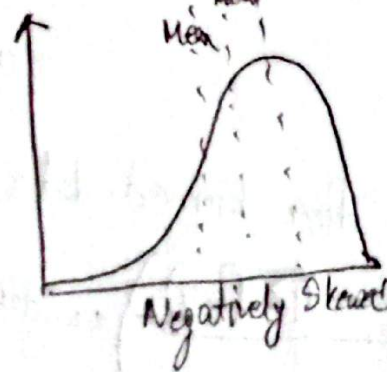
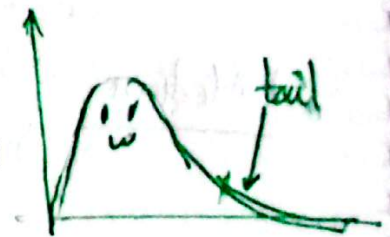
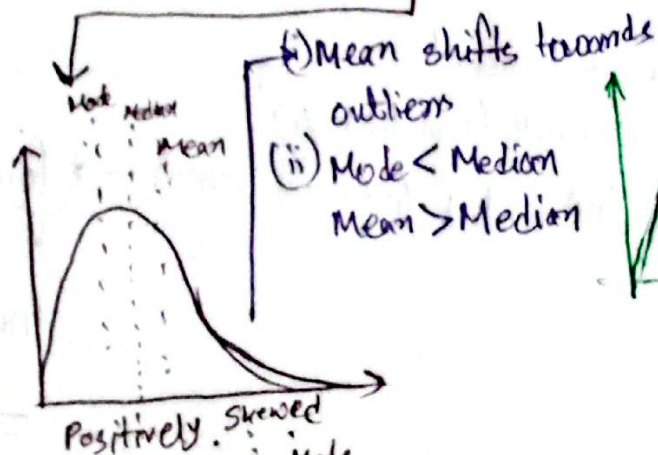
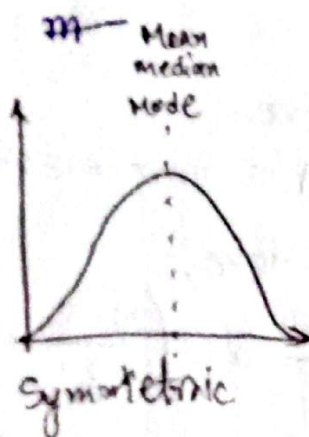
$(\sum \text{freq})_L$ = Cum. frequency upto median interval
width = range of frequency interval

* Mode: Numbers which appears most

→ for unimodal numeric data

moderately skewed (asymmetric)

$$\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$$



C-5, W-3 (Q-2)

08/07/26

Dispersion of Data: -85

*Range $\rightarrow$ Difference betⁿ max and min values

*Quantile $\rightarrow$ Make equal parts ~~between~~ based on data points
after regular interval

$\rightarrow$ 2-Quantile $\rightarrow$ the datapoint that creates two ^{equal} halves of the whole dataset $\rightarrow$ median $\rightarrow$ check the formula for finding quantiles $\rightarrow$ HW
 $\hookrightarrow$ we usually discard the quantile

$\rightarrow$ 4-Quantile $\rightarrow$ the 3 datapoints that create 4 parts
 Q_1, Q_2, Q_3 Quantiles

↓
finds median

↓
median of the two parts

$\rightarrow$ 100-Quantile $\rightarrow$ makes percentile

$\therefore Q_1 = 25^{\text{th}}$ percentile

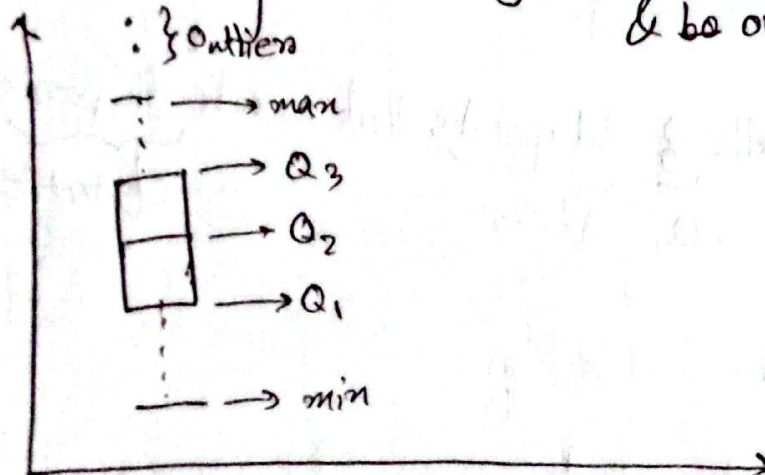
$Q_2 = 50^{\text{th}}$ "

$Q_3 = 75^{\text{th}}$ "

* Interquartile range $\rightarrow$
IQR

- (i) $x \leq Q_1 - (1.5 \text{ IQR}) \rightarrow$ ^{Lower} ~~Small~~ Outlier $\rightarrow$ -ve Outlier
(ii) $x \geq Q_3 + (1.5 \text{ IQR}) \rightarrow$ Large Outlier $\rightarrow$ +ve Outlier

* Box plot: Graphical Representation of using five numbers
(theory summary) $\rightarrow$ min, Q_1 , Q_2 , Q_3 , max
& be outliers



* Variance $\rightarrow$ ^{squared} avg. distance of points from mean
^{magnitude}
^{remove -ve}

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N x_i^2 - \bar{x}^2$$

* Standard Deviation $\rightarrow \sigma$

$\rightarrow$ avg distance . . .

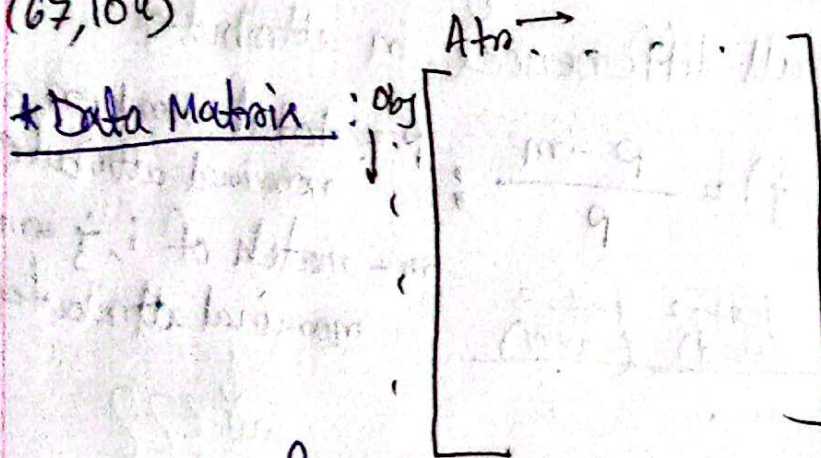
$\rightarrow$ use N for whole population
 $N-1$ " specific part
 $\downarrow$
we do this to avoid bias

C-6, W-3(Q-2)

09/07/26

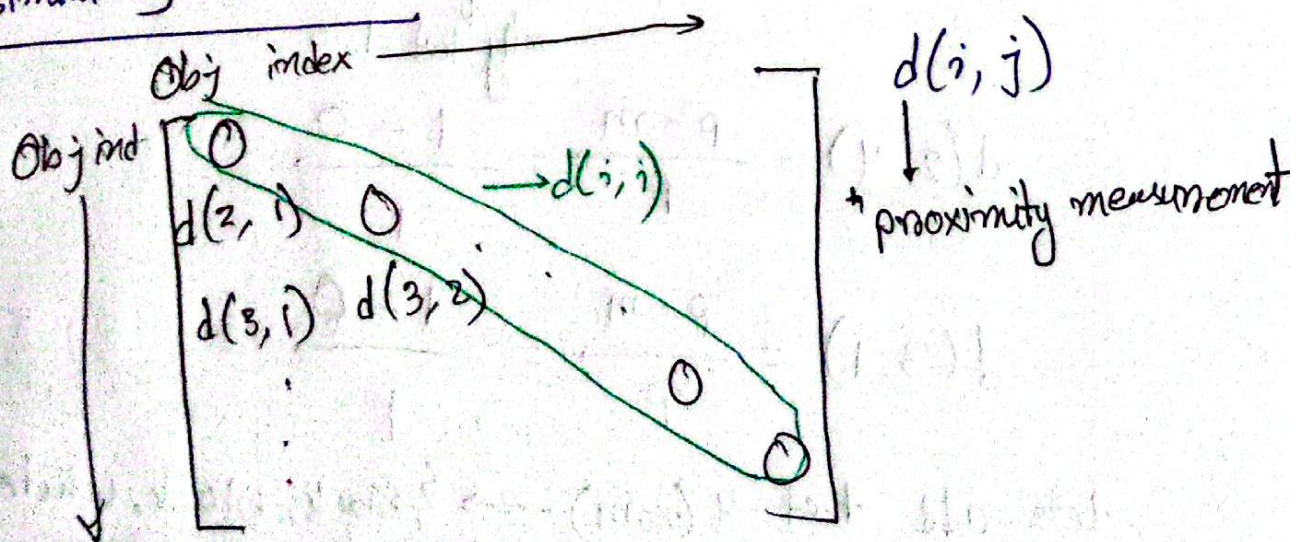
Data Matrix vs Dissimilarity Matrix (2.4.1)

(67, 104)



Collection of
→ Objects in ~~an~~ is data samples or feature vectors,
vectors of attr
of obj

* Dissimilarity Matrix :



Proximity Measure: → differently calculated for different data types

↳ take avg of all differences in attributes

* Nominal: $d(i, j) = \frac{p - m}{p}$; p = total number of nominal attribute
 m = match of i, j among nominal attributes

Say,

Obj identifier	test-1 (nom)	test-2 (ord)	test-3 (num)
1	code A	.	.
2	B	.	.
3	C	.	.
4	A	.	.

wait ???

$$d(1, 2) = \frac{p - m}{p} = \frac{1 - 0}{1}$$

both none are same code (nominal)
 only test-1 is

$$d(2, 4) = \frac{p - m}{p} = \frac{1 - 0}{1}$$

$$d(3, 1) = \frac{p - m}{p} = \frac{1 - 0}{1}$$

lets add test-4 (nom) → {black, black, white, white}

$$d(1, 2) = \frac{2 - 1}{2} = 0.5$$

only black, black are same

* Binary:

→ symmetric: both values are equally important

→ asymmetric: one is more important
 → one is more useful for data analysis
 → the other might be more common

Contingency Table:
 (Symmetric)

	Object j		sum
	1	0	
Object i	1	q	r
	0	s	t
sum	$q+s$	$r+t$	p

q = No. of attr with 1

r = i has 1
 j has 0

s = i has 0
 j " 1

total ~~see~~ symmetric attr.

$$\therefore d(i, j) = \frac{r+s}{r+q+s+t} = \frac{r+s}{p}$$

For Asymmetric:

$$\therefore d(i, j) = \frac{r+s}{q+r+s} = \frac{r+s}{p-t}$$

we ignore 00

$\therefore$ for binary $\rightarrow$ avg (sym, asym)

(online)

C-7 (W-4?) Q-2

15/07/26



* Numeric

→ Euclidean $\rightarrow d(i, j) = \sqrt{(x_{i_1} - x_{j_1})^2 + (x_{i_2} - x_{j_2})^2 + \dots}$

→ Manhattan $\rightarrow d(i, j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + \dots$

→ Minkowski $\rightarrow d(i, j) = \sqrt[h]{|x_{i_1} - x_{j_1}|^h + |x_{i_2} - x_{j_2}|^h + \dots}$

↳ generalizing euclidean & manhattan
usually $h = 1, 2$

→ Supremum $\rightarrow d(i, j)$

↓
the attr before
which dist will
be max

↓
for $x_1(1, 2)$ and $x_2(2, 5)$

$$\therefore \text{supremum} = |2 - 5| = 3$$

* Ordinal → i) String to numeric

ii) Normalize $\rightarrow z_{if} = \frac{x_{if} - 1}{M_f - 1}$; $M_f = \text{max value total number of unique value}$

iii) Use numeric → but for multiple ordinal

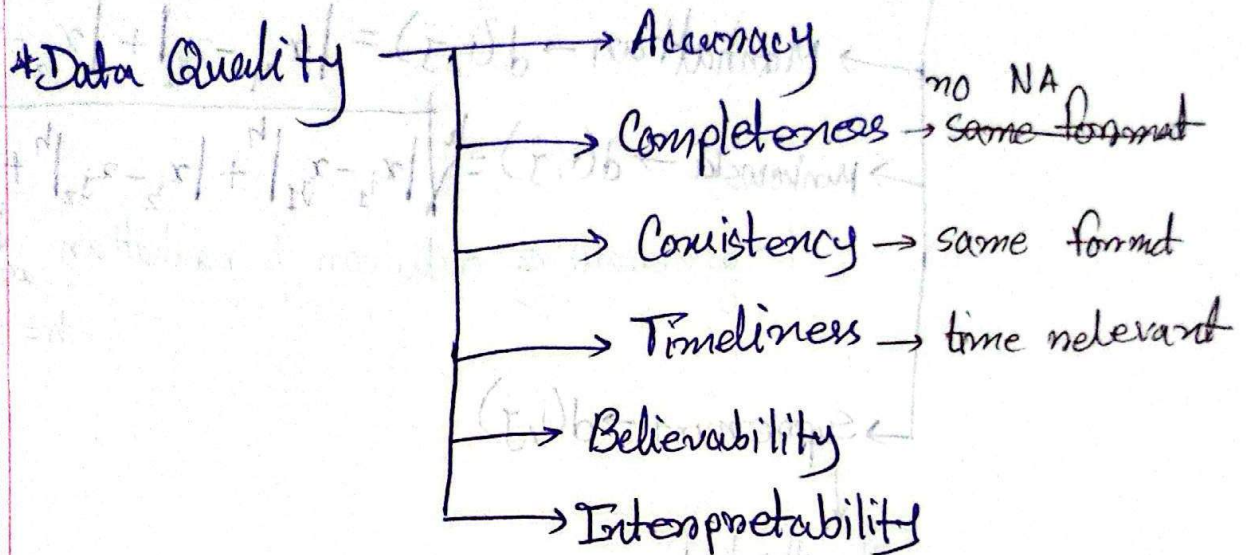
↳ use manhattan minkowski

Quiz - 2 on
week - 6 wed
29th → ch - 2

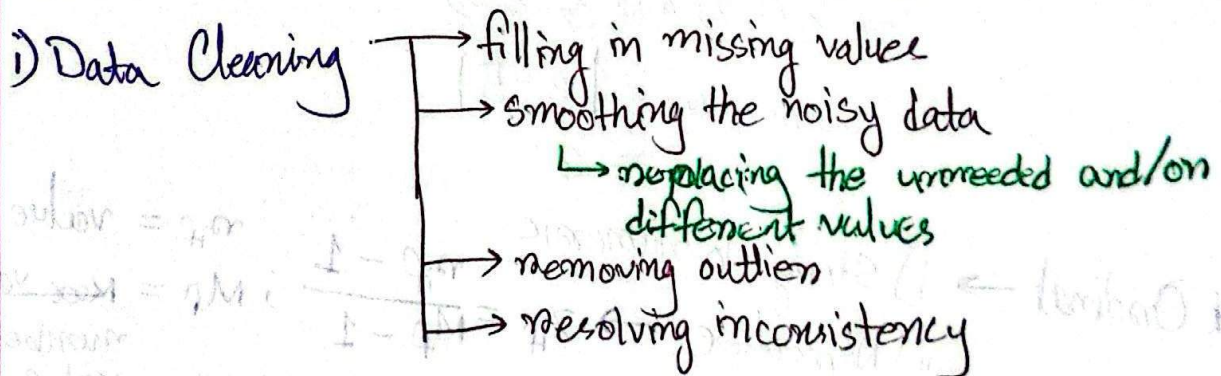
online
C-8, W-4? (a-2.5)

16/07/28

Data Preprocessing

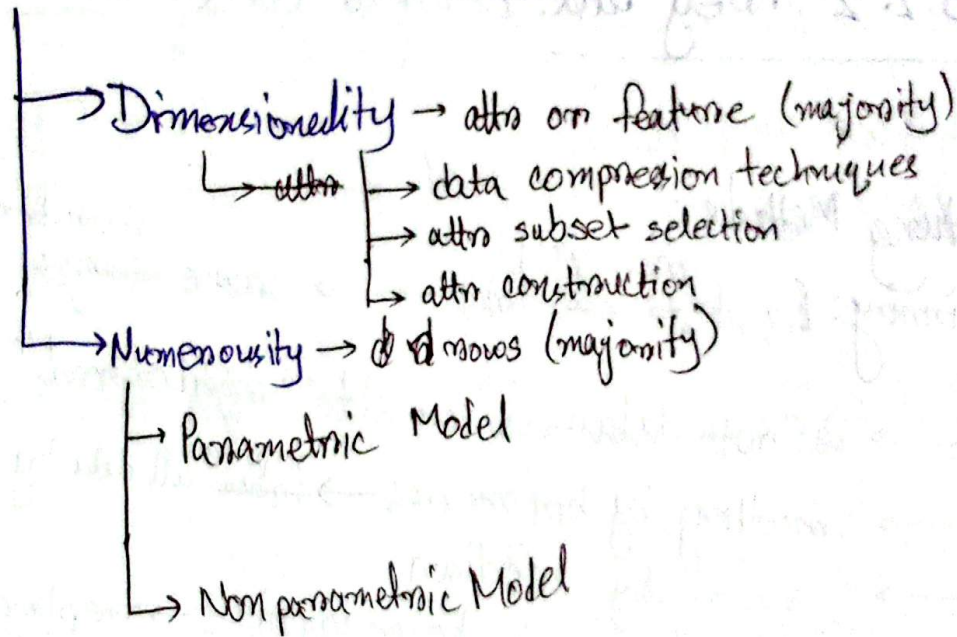


Major Tasks in Data Preprocessing:



ii) Data Integration → merging datasets

iii) Data Transformation Reduction



Data Cleaning:

i) Missing Value:

- i) Ignore the tuple / record ← highly inefficient
 - ii) Filling in manually ← only done when multiple / majority are missing
 - iii) Using a global constant
 - iv) Use mean or median of for all sampling samples belonging to the same class as given tuple
→ find the val not the entire attr. just the ones with same class / label
 - v) Use the most probable value
 - vi) Using mean and median
- Biases the data {

C-9, W-5 (Q-2.5)

20/07/26

3.2.2. Noisy Data: Deviated Data

↳ we smooth it

Smoothing Methods:

i) Binning: Equal ^{frequency} sized bins → same range numbers

- ↳ Sort data → consult "neighborhood"
- ↳ Smoothing by bin means → replace ~~mean~~ all data by the mean *we don't usually replace all values tho*
- ↳ " by median
- ↳ " bin boundaries → replace all data by boundary values

* in exam we replace all values

ii) Regression: Find the "best" possible value to recalculate
Pls don't tell me I need to explain what regression is

iii) Outlier Analysis: [from before]

C-10, W-5 (Q-2.5)

22/07/26

3.3.2. Redundancy and Correlation Analysis:

if two attr are very strongly correlated $\rightarrow$ we will be able to retrieve one from another

* Correlation Analysis $\rightarrow$
 $\rightarrow$ Nominal
 $\rightarrow$ Numerical

Nominal: String / Categorical
 $\rightarrow$ we use χ^2 test (chi square)
 $\rightarrow$ Hypothesis test
 $\rightarrow$ Null hypothesis test
 $\rightarrow$ Assuming no correlation
 $\rightarrow$ freq if not correlated

	Attr. 1		Total
	categ 1	categ 2	
Attr. 2	categ 1 $\rightarrow$ ()	categ 2 $\rightarrow$ ()	Total $\rightarrow$ ()
categ 2	()	()	
Total	()	()	()

$\rightarrow$ expected freq (e_{ij})
 $\rightarrow$ observed freq (o_{ij})

$$e_{ij} = \frac{\text{count}(i) \times \text{count}(j)}{n} ; n = \text{total}$$

$$\chi^2 = \sum \frac{(o_{ij} - e_{ij})^2}{e_{ij}} \rightarrow \text{if } \chi^2 > \text{Table} \rightarrow \text{then NULL Hypothesis then false} \rightarrow \text{correlation exists}$$

$$\text{degree of freedom} = (\text{row} - 1)(\text{col} - 1)$$

significance value $\rightarrow$ given in ques

Deg of freedom	Sig value

C-11, W-5 (Q-2.5)

23/07/26

* Significance Level $\rightarrow$ Prob of making mistake if Null Hypothesis fails

* Correlation $\rightarrow$ Strong if $\chi^2 \gg \text{table}$
Weak if $\chi^2 > \text{table}$

* Deg of Freedom $\rightarrow$ How many values can be independent
 $\rightarrow$ no significance in this regard as we take values from dataset anyways

Numerical: Pearson's Product Moment Coefficient

$\rightarrow$ Covariance $\rightarrow$ take avg $\rightarrow x - \bar{x}, y - \bar{y} \rightarrow$ if same sign then relation
 $\rightarrow$ Correlation Standard Deviation

* Covariance doesn't indicate intensity $\rightarrow$ sign indicates relation
Correlation can but magnitude doesn't mean anything

* We have concepts of r^2 and covariance:

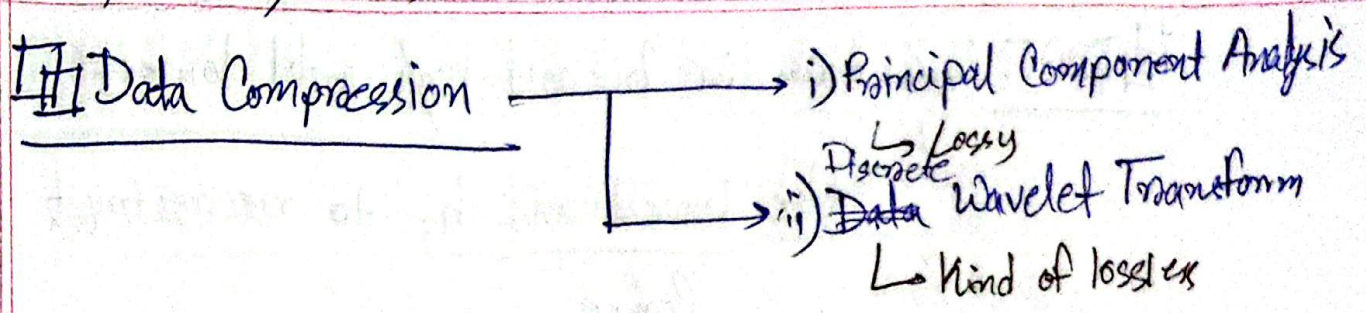
$$\sum \frac{(x - \mu_x)(y - \mu_y)}{\sigma_x \sigma_y}$$

$$\text{Cov} = \sum \frac{(x - \mu_x)(y - \mu_y)}{n \cdot \sigma_x \sigma_y}; \mu_i = \text{avg}(i \text{ col})$$

Q-12, Q-2.5, (w-6)

27/07/26

Data Compression



PCA: Using a single value to express the multiple attribute
 ↳ called Principal Component

* PC1 → Gives highest priority to the attr. with highest variance

↳ $n(\text{Attr.}) = n(\text{PC})$
 but we keep 2
 ↳ but only one is kept

* PC2 → [yeah I think you can make we assign weight corresponding to priority the rest out]

* PC3 → Gives ~~lowest~~ highest priority to the attr. with ~~lowest~~ ^{third} variance

* The more PC, the easier the to retrieve the original data. → but not entirely

* Find λ → Use Eigen vectors → Use the values ^{called Loading} ~~equal~~ in vectors PC
 ↓
 Eigen values
 represents variance

#DWT: Breakdown in low and high until lowest

↳ Use lowest and h_i to reconstruct
Coarse

C-13, W-6, Q-2.5

29/07/26

PCA Math:

<u>Stu Id</u>	<u>Hours Studied</u>	<u>Exam Marks</u>
1	2 (-1.26)	1 (-1.26)
2	3 (0.63)	2 (-0.63)
3	4 (0)	3 (0)
4	5 (0.63)	4 (0.63)
5	6 (1.26)	5 (1.26)

z-score
Step-1: Normalization

$$z = \frac{x - \bar{x}}{\sigma_x}$$

Step-2: i) Covariance using normalization
 ii) " " matrix C

$$\text{CoV} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= 0.992 \approx 1$$

$$C = \begin{bmatrix} \text{Var}(x) & \text{CoV}(x, y) \\ \text{CoV}(y, x) & \text{Var}(y) \end{bmatrix}$$

Step-3: Eigen Values λ

$$C \cdot \vec{v} = \lambda \vec{v} \rightarrow C \vec{v} = \lambda \vec{I} \vec{v}$$

$$\text{or, } C \vec{v} - \lambda \vec{v} = 0$$

$$\therefore \vec{v} (C - \lambda) = 0$$

$$\therefore \text{Either } \vec{v} = 0 \text{ or } C - \lambda \vec{I} = 0$$

But $\vec{v} = 0$ is not possible $\rightarrow$ loadings can never be 0

But $C - \lambda \vec{I} = 0$ is also not possible

but $|C - \lambda \vec{I}| = 0$ is possible

$$\text{or, } \begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$\text{or, } (1-\lambda)^2 - 1 = 0$$

$$\text{or, } 1 - 2\lambda + \lambda^2 - 1 = 0$$

$$\text{or, } \lambda^2 - 2\lambda = 0$$

$$\text{or, } \lambda(\lambda - 2) = 0$$

$$\therefore \boxed{\lambda = 0}, \boxed{\lambda = 2} \rightarrow \text{PC1}$$

$\downarrow$
PC2

Step-4: Eigen Vector

When $\lambda = 2$,

$$\vec{v}(C - \lambda I) = 0$$

$$\text{or, } \vec{v}(C - 2I) = 0$$

$$\text{or, } \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} -v_1 + v_2 & v_2 - v_1 \end{bmatrix} = 0$$

$$\text{or, } \begin{bmatrix} t & t \end{bmatrix} = 0$$

$$\therefore \|\vec{v}\| = \sqrt{2}t$$

$$\therefore \hat{v} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

C-14, W-7, Q-3

06/08/26

DWT Math

i) Set ~~low~~ must be expressible as 2^m

↳ if not, pad with 0s

ii) a) Avg b) Diff of pairs
 ↓ ↓
 ~~high~~ ~~low~~
 low high

iii) Step 2 (ii) of Avg₀ until 1 value

$$\# [8 \ 12 \ 16 \ 40 \ 45 \ 60, 80, 100]$$

$$\text{Avg}_1: \left[\frac{8+12}{2}, \frac{16+40}{2}, \frac{45+60}{2}, \frac{80+100}{2} \right] = [10, 28, 52.5, 90]$$

$$\text{Diff}_1: \left[\frac{8-12}{2}, \frac{16-40}{2}, \frac{45-60}{2}, \frac{80-100}{2} \right] = [-2, -12, -7.5, -10]$$

Again,

$$\text{Ag Avg}_2: \left[\frac{10+28}{2}, \frac{52.5+90}{2} \right] = [19, 71.25]$$

$$\text{Diff}_2: \left[\frac{10-28}{2}, \frac{52.5-90}{2} \right] = [-9, -18.75]$$

Again,

$$\text{Avg}_3 = \left[\frac{17+71.25}{2} \right] = [45.125]$$

$$\text{Diff}_3 = \left[\frac{17-71.25}{2} \right] = [-26.125]$$

$$\therefore \text{Coarse} = 45.125$$

$$\therefore \text{Required Vector} = [45.125, -26.125, -9, -18.75, -2, -12, -7.5, -10]$$

C-15, W-8, Q-3

17/08/26

Attribute Subset Selection: Removing redundant data
(Kind of ~~reduction~~)

- * Forward Elimination
 - * Backward Elimination
 - * Decision Tree Induction
1. Stepwise Forward

2. Stepwise Backward

3. Combination of Stepwise Forward and Backward

* Stepwise Forward:

$$E = \{ \}$$

$$A = \{ A, B, C, D \}$$

Take the highest % of measure from A with E

For ML

~~Insert~~ to E

Loop until combination ~~accuracy~~ measure is lowered

* Stepwise Backward:

$$A = \{A, B, C, D\}$$

Remove element from A and take % measure

Until % decreases

* Combination of Forward and Backward:

Take the Best single } For E
Remove " worst element }

→ This is because if A is Best ACD is ~~worst~~ ^{Best}
Then A can be selected and B can be removed
A will (probably) not be in ~~the~~ subset stepwise
backward

Sampling: Taking part of instances (rows)

→ ~~Random~~ Simple Random Sample Without Replacement (SRS WOR) → single entry once (cut)

→ Simple Random Sample With Replacement (SRSWR)
↳ multiple times (copy)

→ Cluster → Take all from a range → convert that some cluster maybe lost

→ Stratified → Take SRS with ratio to the original dataset

Ch-4: Data Warehouse **

Online Analytical Processing → OLAP → DW
" Transactional " → OLTP → DB

* Data Warehouse is made using Database using Aggregation
* DB has details | Data Warehouse has compressed details of DB

* DW is used to analyze conditions

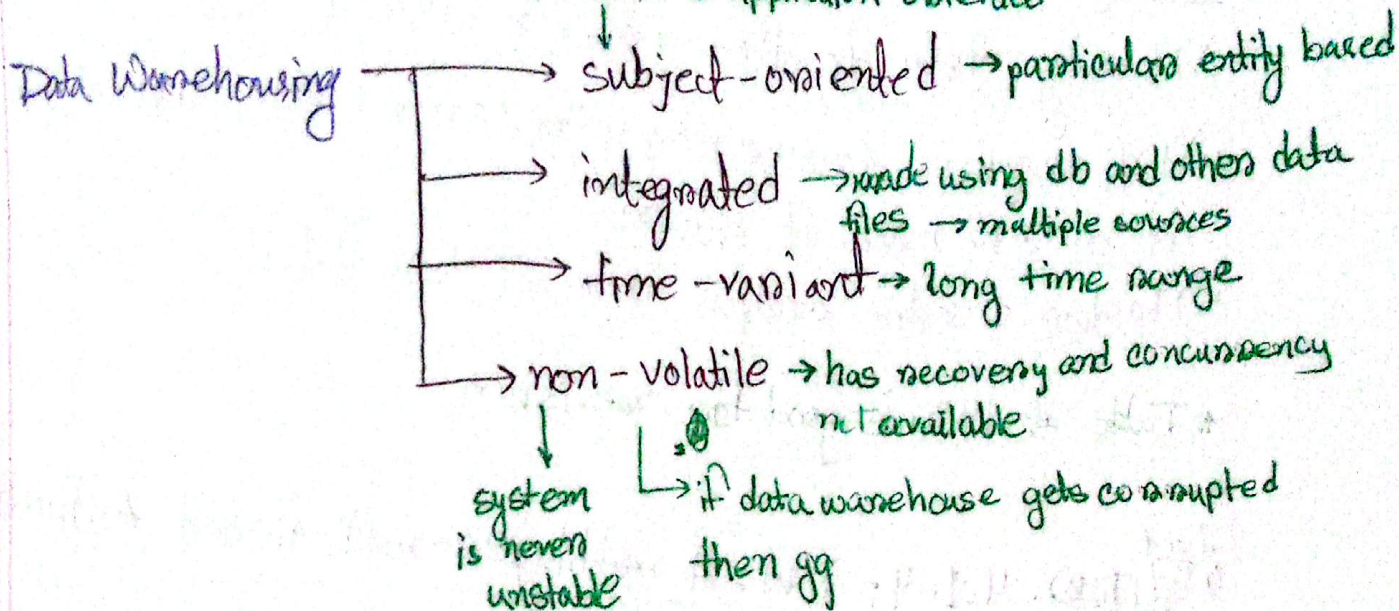
DB has recent but small data | DW has overall (old) but huge data

(16: Compiler)
C-17, W-9, Q-3.5

24/08/26

Data Warehousing

Database is application oriented



Differences between DB (OLTP) and DW (OLAP):

	customer oriented	market oriented
i) users and system orientation		
ii) Data contents	Too detailed for decision-making	Short but huge and good for decision-making
iii) Design	Entity-Relation (Table)	Different schemes (Star, Snowflake, Constellation and Data cube)
iv) view	Current	Historic
v) Access Pattern	Read & write	Read

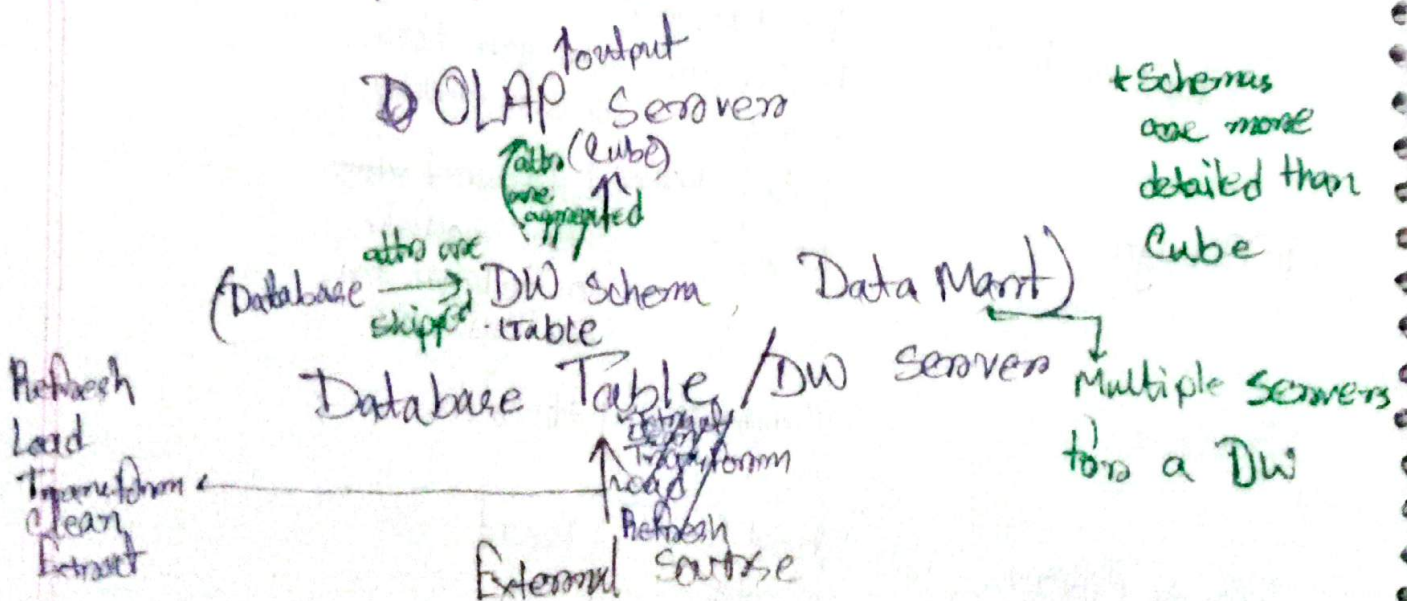
27/08/26

- i) Merging them together makes it slower
- ii) DB requires concurrency, recovery etc and implementing it in DW makes it slower
- iii) Faster market analysis requires less details

* Table at 130 is good for revision

+ 3 tiers (but can be more)
↳ Most popular one

Frontend Tools



C-19, W-10, Q-3. 5

31/08/26

- + Enterprise Warehouse: Single admin panel
- * Data Mart: Diff. DW, ^{schemas} for diff case
- * Virtual Warehouse: Temporary schema that is deleted after use

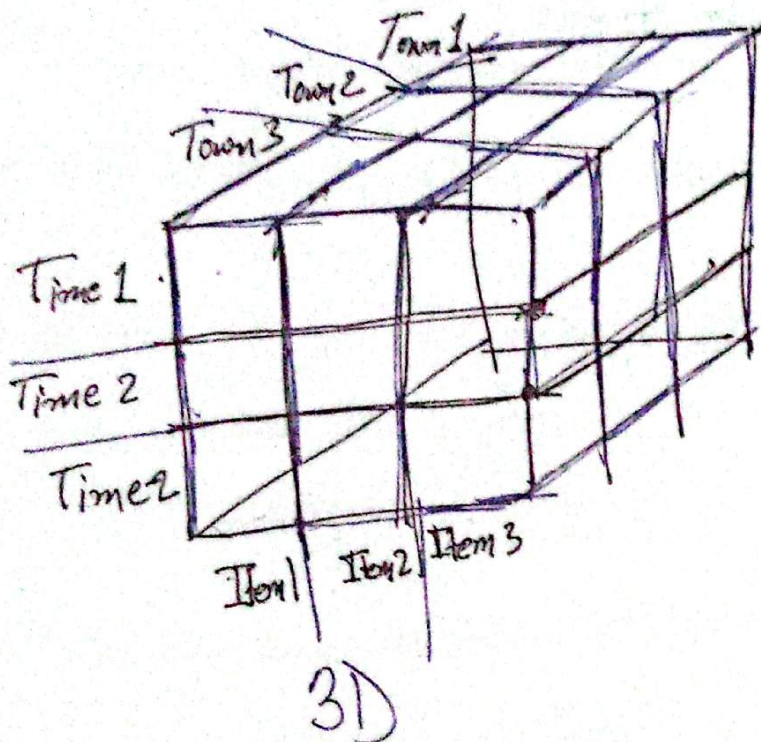
Discuss the recommended approach.

C-20, W-10, Q-4

3/9/26

4.2.1. Data Cube: A Multidimensional Data Model

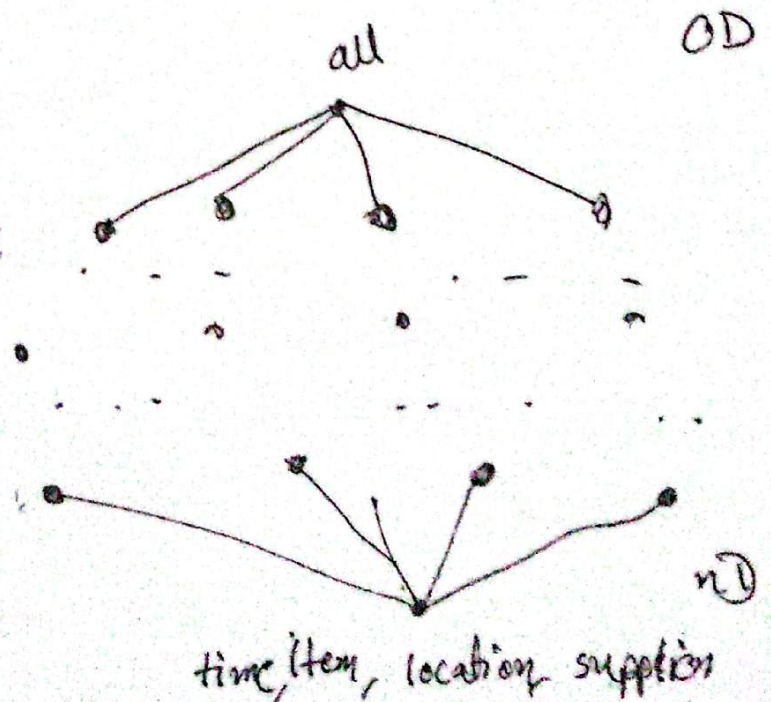
* Highest priority is given to dimension



~~Lattice~~

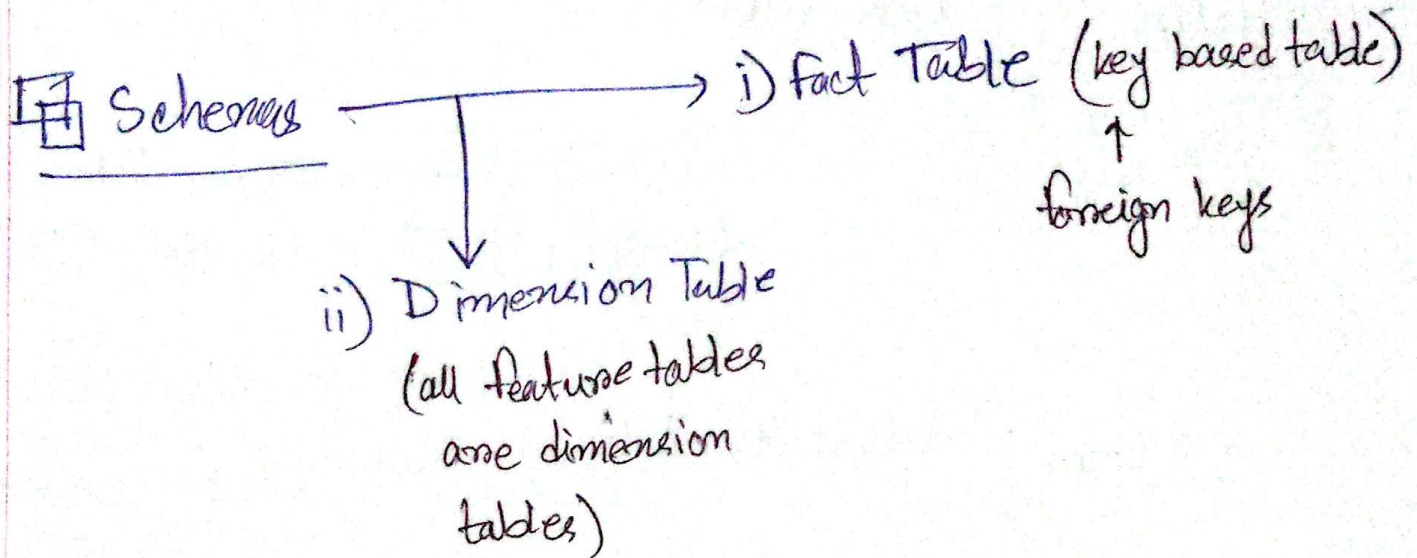
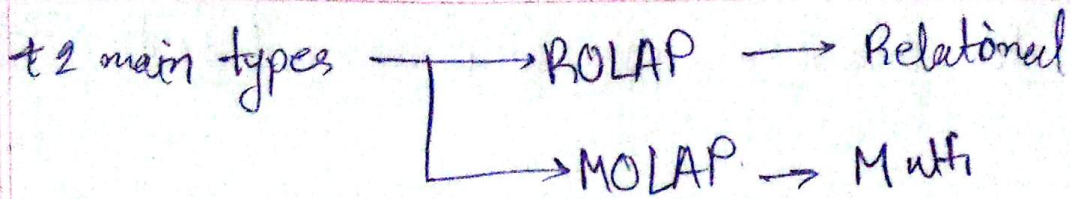
* Lattice of Cuboids:

↑ aggregates
because
dimension
reduces



C-21, W-11, Q-A

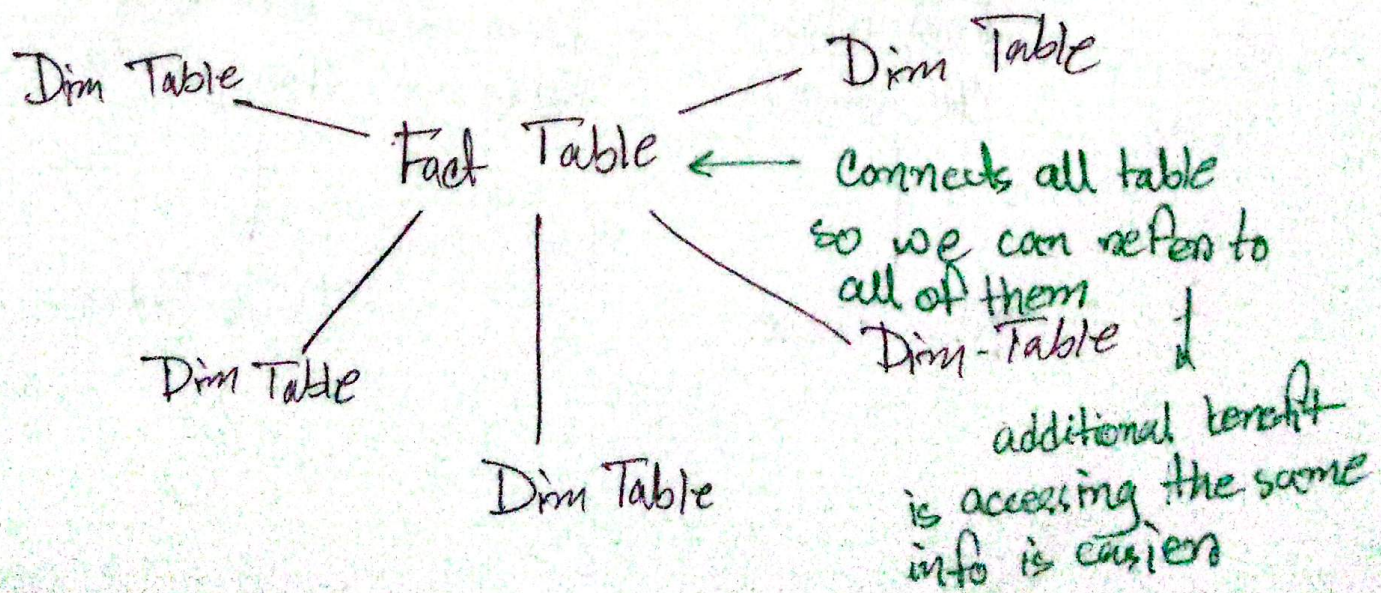
10/09/26



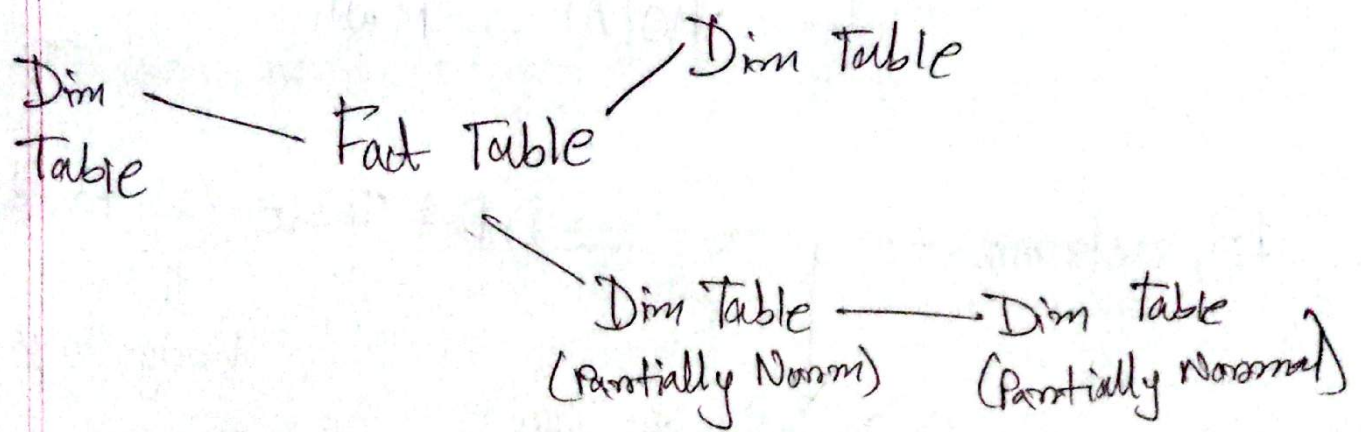
*Star Schema: Only a single fact table

DB ↓

↳ Dimension tables are not normalized



* Snowflake: Stars but partially normalized

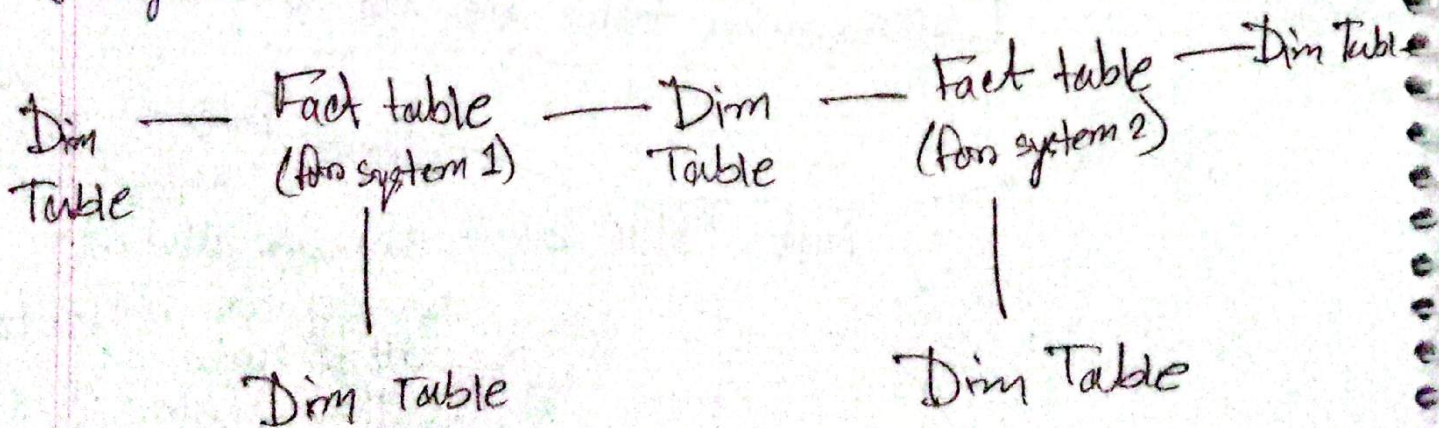


→ Stars is better for DW

* Fact Constellation: Multiple Fact Tables

aka
Galaxy Schema

↳ For Multiple Systems



4.2.5. Typical OLAP Operation

* we access basic cube from ~~in front~~ using frontend
↳ Only partial/as much needed

* roll-up: hierarchy upping (cities → country)

* drill-down: hierarchy drilling (country → cities)

* slice: selecting specific attr.

* dice: selecting multiple attr. with multiple values

* pivot: rotates / interchanges data

C-22, W-12, Q-1

14/09/26

Aggregate Function

i) Distributive: Applying full set = ~~ap~~ Applying on parts
(Sum)

ii) Algebraic: A Combination of multiple distributive
(Average)

iii) Holistic: Storage size doesn't have upper bound
(median)

↓
Higher numbers goes beyond computers
processing/memory size

4.3.2. Data Warehouse Design Process:

→ Top-Down → make enterprise first
and then data marts

Bottom
Down - Up → make data marts
then ensemble enterprise

no. of
subclasses of i th class

$$\boxed{\text{Number of Cuboids}} = \prod_{i=1}^n (L_i + 1) \quad \begin{array}{l} \text{for subclasses} \\ \text{all} \\ \text{for no subclasses} \end{array}$$

2^n

C-23, W-12, Q-A

15/09/26

Materialization: Perspective/Slices of combination

↳ saves time but takes storage

i) No Materialization: We do not precompute any cuboid

ii) Full Materialization: All subcubes precompute

iii) Partial Materialization: Only some are precalculated

↳ The frequent ones

Indexing:

i) Bitmap Indexing: Make columns with 0/1 instead of going through the full table

ii) Join Indexing: Materialize Index Tables $\rightarrow$ Fact Entries

C-24, W-12, Q-f

17/09/26

Mining Frequent Patterns, Associations and Correlations

Using algorithms

Can have Cond
but A has to be on LHS
and B has to
be on RHS

$$* \text{support}(A \Rightarrow B) = P(A \cup B) = \frac{\text{count}(A \cup B)}{\text{Total no. of transactions}}$$

$$\text{confidence}(A \Rightarrow B) = P(A|B) = \frac{\text{count}(A \cup B)}{\text{count}(A)}$$

□ Closed Frequent Itemset
& Maximal " "

} we don't usually consider
high weak / less frequent
as items

* all the subsets of a frequent itemset are also frequent

Keeping all freq itemset are storage heavy

'We keep some → two universal rules

→ Closed
→ Maximal

* Closed Frequent Itemset: Only keep the ones that satisfies
the threshold $\rightarrow$ discard the rest

if no sup proper superset $\rightarrow$ count it
 $\rightarrow$ else $\rightarrow$ if $\text{superset}(\text{count}) \neq \text{set}(\text{count})$
 $\rightarrow$ count it

* Maximal Frequent Itemset: Only keep the ones that
satisfies the threshold

$\downarrow$
if (! proper superset)

$\downarrow$ not
if proper superset is frequent

$$\begin{aligned} & \# \{a_1, a_2, \dots, a_{100}\} : 1 \\ & \{a_1, a_2, \dots, a_{50}\} : 2 \end{aligned} \quad \left. \vphantom{\begin{aligned} & \# \{a_1, a_2, \dots, a_{100}\} : 1 \\ & \{a_1, a_2, \dots, a_{50}\} : 2 \end{aligned}} \right\} \text{Threshold} = 1$$

Closed $\rightarrow$ Both

Maximal $\rightarrow$ Only first